MILITARY SPECIFICATION SHEET

PAINT, EPOXY-POLYAMIDE, EXTERIOR TOPCOAT, HAZE GRAY, FORMULA 151, TYPE I

This specification is approved for use within the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the paint described herein shall consist of this document and the latest issue of MIL-P-24441(SH).

FORMULA. This formula covers haze gray (R=27) epoxy-polyamide paint designated Navy formula 151, type I, for interior or exterior use. The paint shall consist of the ingredients specified in the quantities specified.

Component A

	Pounds
Thixotrope[1/]	-
Polyamide[2/]	20
Polyamide adduct[3/]	280
Magnesium silicate[4/]	375
Titanium dioxide[5/]	100
Butyl alcohol[6/]	293
Lampblack[7/]	1.5

Component B

	Pounds
Thixotrope[1/]	-
Epoxy resin[8/]	500
Magnesium silicate[4/]	250
Naphtha[9/]	251

See footnotes on next page.

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QUANTITATIVE REQUIREMENTS. The paint shall meet the following quantitative requirements and the qualitative requirements of section 3 of the general specification. Components A and B shall be mixed 1:1 by volume for mixed components tests. Tests shall be performed in accordance with the general specification.

Requirements	Component A		Component B		Mixed component	
	Min	Max	Min	Max	Min	Max
Pigment content, percent	43.0	47.0	23.5	27.5	---	---
Volatiles, percent	30.5	34.5	22.8	26.8	---	---
Nonvolatile vehicle, percent	19.5	23.5	47.5	51.5	---	---
Water, percent	---	1.5	---	0.5	---	---
Coarse particles, percent	---	0.3	---	0.3	---	0.5
Consistency, grams	165	310	115	185	120	170
Pounds per gallon	10.4	10.9	9.8	10.3	10.1	10.6
Set-to-touch time, hours at 40°F	---	---	---	---	---	2
Set-to-touch time, hours at 73°F	---	---	---	---	---	2
Dry-hard, hours at 40°F	---	---	---	---	---	24
Dry-hard, hours at 73°F	---	---	---	---	---	6
Fineness of grind, NS	5	---	5	---	5	---
Flash point, °F	99	---	115	---	99	---
Titanium dioxide, percent of pigment	18.5	---	---	---	---	---
Pot life, hours at 73°F	---	---	---	---	6	---
Gloss, 60° specular, percent	---	---	---	---	---	30
Sag resistance, mils	---	---	---	---	8	---
Color dry film to match standard color chip 10/	---	---	---	---	Conform	
Weight per epoxy, (vehicle)	---	---	175	195	---	---
Contrast ratio, 3 mils dry film	---	---	---	---	0.98	---

1/ The amount and type of thixotrope shall be selected by the manufacturer to meet all requirements of the general specification and this specification sheet.
2/ GENAMID 2000, Henkel Corporation; UNIREZ 2810, Union Camp; AZAMIDE 600, AZ Products; ANCAMINE 507, Pacific Anchor Chemical Corp.; EPOTUF SF7791, Reichhold Chemical; TriChem 9200, Trimont Chemicals.
3/ VERSAMID 280B75, Henkel Corporation; UNIREZ 2180B75, Union Camp; AZAMIDE 680B75, AZ Products; ANCAMINE 700B75, Pacific Anchor Chemical Corp.; EPOTUF SF7792, Reichhold Chemical; TriChem 9280-B-75, Trimont Chemicals.
4/ Pfizer Talc MP 30-36; Cyprus Montana Talc, Mistron 400.
5/ DuPont Tipure R960; NL Chemicals, TX 2101; Kerr-McGee, Tronox Tr 822.
6/ Conforming to ASTM D 304.
7/ Pfizer M1011.

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8/ EPON 828, Shell Chemical Co.; ARALDITE 6010, Ciba-Geigy; DER 331, Dow Chemical Co.; AZEPOXY 128, AZ Products; EPO-TUF 37-140, Reichhold Chemical; EPI-REZ 510, Celanese; TriChem 728, Trimont Chemicals.

9/ Amsco Super High Flash Naphtha.

10/ Use FED-STD-595 Color Chip-Color No. 26270. When the instrument method of measuring color difference is used the color difference (E) shall be no more than 4 units.

Use of alternate ingredients in this formula must have prior approval by the Naval Sea Systems Command. Approval will be based on review of data showing equivalent physical and chemical characteristics to the specified ingredient. It will be necessary to demonstrate that paint made using the ingredient will conform to all requirements of this specification.

Revision letters are not used to denote changes due to the extensiveness of the changes.

Preparing activity:
Navy - SH
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